

Liqo: Reversible Commitment in LLM Exploration through Liquid-Solid Interaction in VR

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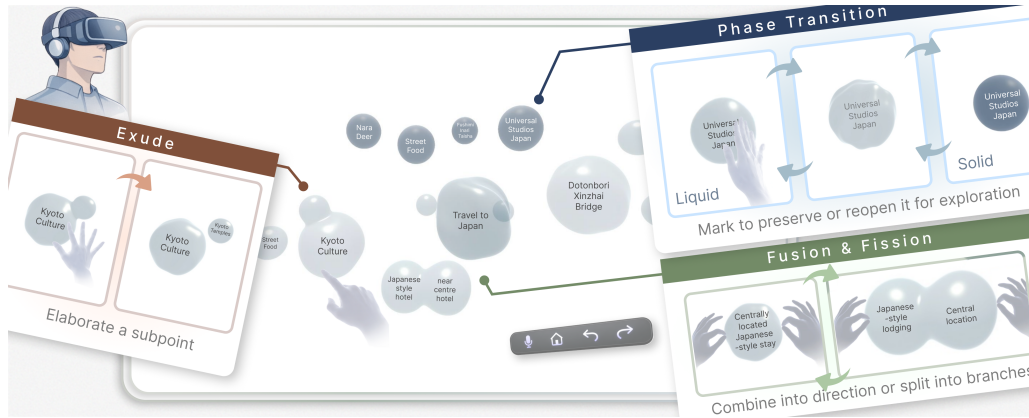


Fig. 1. Liqo is a VR interface for exploratory human-LLM sensemaking built around a liquid-to-solid phase-transition metaphor. AI-generated information begins as liquid bubbles for low-commitment exploration, and can be frozen into solid orbs for more stable organization and synthesis. Users can later melt solid orbs back into liquid form to reopen and revise earlier decisions across a spatially organized 3D workspace.

Exploratory work with large language models requires stabilizing tentative material while keeping it available for revision. We present Liqo, a virtual reality interface in which users compose liquid information bubbles, freeze selected material into solid orbs, and melt them to resume exploration. A within-participant study with 13 participants compared four interfaces across three planning tasks with changing requirements. Liqo met all critical task criteria in 97.4% of trials versus 61.5% for the VR baseline, with shorter restricted time to the first recorded passing state. The larger endpoint difference concerned maintaining or completing an executable proposal, alongside higher same-key content-and-lineage continuity. Cases show that reopening an object can precede content correction and that critical task success can coexist with discontinuity. These findings inform workspace designs that make working status, content updates, and task checks separately inspectable.

CCS Concepts: • **Human-centered computing** → **Systems and tools for interaction design**.

Additional Key Words and Phrases: Human-AI Interaction, Exploratory Interaction, Large Language Models, Sensemaking

1 Introduction

A plan produced with a large language model (LLM) is often useful before it is finished. A traveler may settle on an accommodation area while still comparing activities; an event organizer may preserve a schedule while reconsidering a venue. When a budget changes, revision involves identifying affected decisions, recovering alternatives, and retaining work that remains useful. The interface must support both a stable working representation and a way to reopen it. Sensemaking research describes this tension: people construct provisional representations to reason with, then revise those representations as evidence and goals evolve [12, 24, 25].

Conversational interfaces provide an accessible way to express changing goals, but successive responses can separate a decision from its alternatives and assumptions. Spatial LLM systems address parts of this problem through persistent nodes, diagrams, and branching workspaces. Sensecape supports multilevel exploration [31], Graphologue makes generated relationships interactive [9], and Luminate organizes alternatives [30]. Once users select material within such a workspace, a further interaction question arises: how can treating an idea as *settled for now*, and later reopening it, become an explicit part of manipulating that idea?

We use *reversible commitment* to mean marking material as settled for now and returning it to an editable working state. This operational definition concerns a user’s control over the workspace, without inferring confidence, certainty, or factual verification. Pinning, draft/final markers, and history controls already support aspects of this goal. We investigate how a material vocabulary can coordinate a legible working state, operations appropriate to that state, and continuity between reopened material and its content and derivation. The design challenge is to preserve a usable reference without making a provisional choice difficult to revise.

We present *Liqo*, a virtual reality (VR) interface that realizes these transitions through a liquid–solid mapping (Figure 1). Liquid bubbles support gathering, fusion, fission, and local elaboration. Freezing marks selected material as solid orbs for stable arrangement; melting reopens those objects for exploration in the current workspace. Content and derivation links accompany the representations. The design contribution lies in coordinating familiar composition and history functions through object-level working states. This mapping is selective: solid form expresses a user’s selection, while the correctness of generated content still requires inspection. Research on AI metaphors motivates making such correspondences explicit [22, 23, 29].

VR provides a setting for two-handed composition and separation, with surrounding space for active material, staged alternatives, and stabilized structures [11, 13, 19]. We developed the design through a synthesis of formative input from ten frequent LLM users, prior work, and design reasoning. An exploratory evaluation then compared four interfaces across 156 planning trials from 13 participants. *Liqo* and a VR baseline share a model and base prompt, allowing a focal comparison of the tested immersive configurations. Desktop conditions provide context but use a different model and prompt.

After requirements changed, *Liqo* achieved higher final critical-constraint success and shorter restricted time to the first recorded passing state than the VR baseline. A descriptive decomposition locates the larger endpoint difference in maintaining or completing an executable proposal, alongside a difference in satisfying changed requirements. *Liqo* also had higher content-and-lineage continuity among still-applicable units assessed under the same structured keys. Cases explain why these outcomes must be considered together: an object can reopen before its content is corrected, and a plan can pass critical criteria while losing continuity for a useful alternative. These findings concern the implemented configuration and bounded planning tasks; they do not isolate the effects of the metaphor or immersion.

Our contributions are:

- (1) *Liqo*, a VR interaction design that coordinates composition, local elaboration, stabilization, and object-specific reopening through a liquid–solid mapping.
- (2) Comparative evidence that the tested *Liqo* configuration supports whole-plan revision, with an advantage involving both changed requirements and the maintenance or completion of an executable proposal, accompanied by higher same-key continuity and lower destructive editing.
- (3) Case-grounded guidance for reopenable LLM workspaces: expose working status, content updates, and task checks separately, and evaluate useful content continuity alongside task success.

2 Related Work

2.1 Metaphors for AI Interaction

Metaphors influence which properties of an unfamiliar system become visible and which expectations seem reasonable. Murray-Rust et al. describe how metaphors for AI can illuminate one aspect of a system while hiding another [23]. Their later work on *metaphor gardening* advocates precise, plural, and intentional use of metaphors, illustrated through design trajectories in which metaphors shape prototyping [22]. These accounts motivate treating a metaphor as a design choice whose consequences require examination, rather than as a natural description of AI.

Desai and Twidale show that metaphors for voice interfaces are contextually fluid and may differ among designers, users, manufacturers, and researchers [5]. A designer’s intended mapping therefore cannot substitute for evidence of a user’s interpretation. So et al. reposition anthropomorphism as a design variable and propose a spectrum of LLM interface metaphors that foreground material and sociotechnical conditions [29]. Their objective is critical engagement rather than simply optimizing usability. Liqo addresses a narrower operational question: how material states can express a user’s provisional treatment of generated information. It does not test whether material imagery reduces anthropomorphism or increases awareness of an LLM’s infrastructure.

Material approaches are already part of LLM interface research. Clay Play frames dynamic idea generation and exploration through a metaphorical LLM interface [1]; Texterial develops a text-as-material interaction paradigm [26]. Liqo extends this line of inquiry to the relation between provisional and stabilized material. Its liquid–solid mapping integrates bidirectional state changes with composition and local elaboration, raising the question of how working status organizes action and continued access to prior work.

Metaphor highlights selected relationships without reproducing an entire source domain [14]. For an LLM workspace, this makes the connection between visible transformations and content changes a design question: a familiar material action can set expectations that generation does not fulfill. Sections 3 and 4 specify Liqo’s mapping and operational limits.

2.2 Spatial and Direct Manipulation Interfaces for LLMs

Sensecape combines a hierarchical canvas with multilevel exploration [31]. Its within-participant study found that externalizing levels of abstraction supported exploration of more topics and hierarchical knowledge organization. Graphologue’s formative study identified difficulty comprehending and flexibly interacting with long responses; its interface turns entities and relationships in those responses into interactive diagrams [9]. Together, these studies connect the representation of generated information to how people navigate and organize it. ExploreLLM structures exploration into subtasks [16], while Luminate organizes alternatives for divergent exploration [30]. These systems provide foundations for persistent, revisitable LLM workspaces.

Revising a travel plan in such a workspace can involve returning to a branch, changing its content, and updating relationships. Liqo makes a further distinction within that workflow: selected material has a working status that controls its availability for composition and elaboration. Reopening a stabilized object changes how the user can act on it while retaining a reference within the current workspace. The focus is the coordination of representation, available action, and provisional commitment.

DirectGPT situates LLM operations on selected text [21], continuing the direct-manipulation tradition of acting on visible objects [28]. Texterial makes generated text available for material-inspired interaction [26]. Liqo builds on

Table 1. Liko’s positioning by design emphasis. The table compares research questions, not the absence of capabilities in prior systems.

Lineage	Shared foundation	Liko’s focus
Sensecape; Graphologue	Persistent units, branching, and relationships.	Expose treating material as settled and reopening it as transitions on that material.
DirectGPT; Texterial; Clay Play	Content-oriented or material-inspired interaction.	Coordinate composition and elaboration with bidirectional stabilization.
Spatial hypertext; ImaginationVellum	Emergent organization, context, and history.	Keep reopened material connected to its content and derivation in the current workspace.
VIVRA; LitForager	Immersive information workspaces.	Enact these transitions through bodily action and spatial staging.

these ideas by connecting content transformations to user-selected states. Fusion and fission remain familiar forms of composition and differentiation; the design contribution concerns their coordination with stabilization and reopening.

Spatial hypertext provides an earlier foundation. VIKI and subsequent work support emergent relationships through spatial organization before users formalize a structure [20, 27]. ImaginationVellum uses the canvas as a spatial prompt and retains ideation history [18], while Garden of Papers brings finding, reading, and organizing literature into one workspace [15]. These systems make context and continuity central to interaction. Table 1 locates Liko’s emphasis within this lineage.

This lineage suggests evaluating both access to earlier material and the content available after revision. Liko’s evaluation distinguishes restoring the ability to act on an object, maintaining structured content and lineage, and reaching a plan that passes task criteria. Considering these outcomes together connects state control to the consequences of ongoing generation.

2.3 Sensemaking and Its Embodied Realization

Sensemaking theories describe the construction and revision of representations, rather than a fixed progression from question to answer. Pirolli and Card distinguish interconnected foraging and sensemaking activities [24]; Russell et al. emphasize the costs of finding and replacing representations [25]; Data/Frame theory explains how evidence can elaborate, challenge, or replace an existing frame [12]. Berry picking and nonlinear information-seeking models likewise situate revision within exploration [3, 7]. These accounts motivate reopenable working structures without implying that human reasoning has only two states.

Studies of desk organization and spatial action show how arrangement can support ongoing work without formalizing every relationship [10, 17]. Embodied and distributed accounts connect reasoning to action, tools, and surroundings [6, 8, 11, 13]. Immersive analytics investigates these opportunities in three dimensions [4, 19], while large-display research describes spatial support for parallel sensemaking [2]. VIVRA and LitForager bring LLM-supported idea exploration and literature foraging into VR [32, 33].

These accounts inform different parts of Liko’s design: emergent structure motivates movable provisional units, reframing motivates object-specific reopening, and spatial action motivates staging alternatives around active work. VR makes those operations physically enactable, while adding access and physical costs that the evaluation examines alongside revision outcomes.

3 Design Grounding and Rationale

Liqo addresses four requirements for working with provisional information. We derived them through an author synthesis of formative input, the affordances of a canvas probe, and prior work on emergent organization and reframing. The requirements motivate the design; the liquid–solid realization is our design choice.

3.1 Formative Design Input

Ten frequent LLM users from a local university participated through convenience sampling. All were STEM students, with a mean age of 20.0 years (SD 1.05; four women and six men; Appendix Table C.1). The study underwent the applicable institutional ethics review, and participants consented to participation and recording. A 20-minute retrospective interview asked about a recent exploratory AI-supported task, focusing on storing, revisiting, comparing, and manipulating generated information. Participants then used a canvas probe for approximately 20 minutes to plan a seven-day trip to Japan’s Kansai region.

The probe ran on a Microsoft Surface Studio with pen and touch input and used Qwen-Plus. Persistent cards supported dragging, splitting, grouping, linking, local follow-up queries, saving, and report construction (Appendix A). Participants thought aloud while screen activity and interaction logs were recorded. Accommodation, activities, transport, and pacing made dependencies among choices concrete. The account describes intermediate organization and revision, but its analysis and observation-to-requirement chain are incompletely documented. We therefore use it as situated design input rather than an independently established set of qualitative findings.

3.2 Requirements and Design Choices

DG1: Support low-commitment reorganization. The formative account describes gathering and rearranging material before establishing a final itinerary. The probe offered dragging and grouping without requiring every card to occupy a formal role. In spatial hypertext, such informal arrangement can precede explicit structure [20, 27]. We carried this possibility into movable liquid units: alternatives can remain in the workspace while users decide how to organize them. The requirement is to externalize work without making early organization costly to change.

DG2: Pair compact units with selective local expansion. An itinerary needs both an overview and details about each choice. Cards, local follow-up queries, and the probe’s response panel exposed these complementary levels of work. Liqo uses compact labels as handles for richer drafts, and exudation to develop a subtopic while keeping its parent as an organizing reference. Fission instead foregrounds separate branches. Preserving a parent favors local continuity, while branching gives alternatives more independent roles; neither operation removes the need to inspect the resulting text.

DG3: Make provisional commitment legible. The probe’s saved-card area and report construction distinguished selected material from active work. Liqo expresses this distinction on the object itself, so stabilized choices and provisional alternatives can coexist in the same workspace. Solid form marks a selection and limits further structural transformations until the object is reopened. This makes working status visible but introduces a deliberate extra transition before further composition. The design trades immediate editability for an explicit distinction between current selections and material still being developed.

DG4: Reopen selected material with continuity. Returning to earlier alternatives connects the formative setting to reframing in sensemaking [12]. A changed walking limit, for example, can require revising part of an itinerary while other choices remain usable. Liqo therefore reopens an individual object in place, retaining its identity and content

references. Source relations record how material was derived. These mechanisms supply a continuing target for work; evaluating whether useful content survives subsequent generation requires a separate content measure.

3.3 Choosing a Material Mapping

Liquid and solid connect these requirements through a common vocabulary. Gathering, combining, and separating liquid units express provisional organization (DG1); attached child material expresses local elaboration (DG2). Freezing changes the appearance and available actions of selected material (DG3), and melting restores its availability for exploration (DG4). The same object can participate in both roles, avoiding a mandatory transfer into a separate final-document interface whenever its working status changes.

Pinning, status markers, layered canvases, and object histories provide alternative realizations. We chose the liquid-solid mapping for its ability to connect composition and stabilization through material transformations, rather than through an additional status vocabulary. We did not systematically compare candidate metaphors. Its intended coherence also creates risks: solidity can suggest correctness, and melting can suggest restoration of an earlier meaning. We define solidity as user-selected working status and melting as renewed manipulability; semantic changes and task validity remain separate. This selective correspondence follows accounts of intentional AI metaphor design [22, 23].

We implemented the mapping in VR to explore two-handed manipulation and spatial staging. Reaching and turning support arrangement but also introduce navigation, reading, and physical costs. The requirements apply more broadly to reopenable workspaces; the evaluation concerns this immersive realization.

4 Liko: Reversible Commitment in a Material Workspace

Liko represents generated information as objects whose appearance and available operations reflect a user-selected working status. A compact label supports spatial organization, while a longer draft carries the proposal, assumptions, and tradeoffs. The interface combines direct manipulation with natural-language input so users can organize a plan and revise its details within the same workspace.

4.1 Working with a Plan

A user begins with typed or spoken intent, such as a travel goal and its constraints. Generated topics appear as liquid bubbles. Grabbing an object makes its draft readable; the label remains a handle for that richer content. A local question or correction supplies explicit intent together with target content to a model request. Users can therefore specify a numerical walking limit or ask for a comparison in language while manipulating the objects that organize the work.

The workspace offers three regions. The near field holds active material within reach; the mid field stages relevant alternatives outside immediate focus; the far-field output wall holds stabilized objects arranged into a plan or outline. These regions scaffold organization without assigning psychological meaning to distance. Users may arrange and inspect objects in either material state.

As a proposal develops, users compose and elaborate liquid material, freeze selected objects for stable arrangement, and continue exploring other branches. A changed requirement can prompt them to melt a selected object and resume work on it while leaving the surrounding arrangement in place. Inspection remains necessary after a content update: changing an object's working state does not itself correct a budget or revise an itinerary. Figure 2 illustrates the overall workflow; Figure 3 summarizes the five material operations.

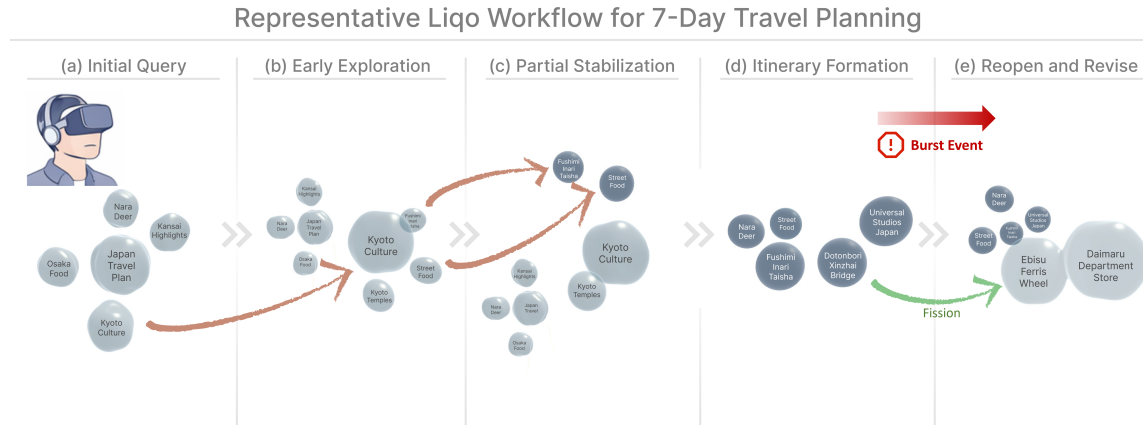


Fig. 2. Illustrative Liqo workflow for seven-day travel planning. (a) An initial query produces broad liquid topics. (b) Exploration develops local branches. (c) Selected material is frozen while other branches remain liquid. (d) Stabilized objects form an itinerary. (e) A changed requirement prompts reopening and reorganization. This design illustration is distinct from the three-day evaluation task.

4.2 States and Transformations

Figure 3 presents five material operations. *Fusion* combines liquid objects to bring related ideas together. *Fission* separates a liquid object into branches. *Freeze* changes a liquid object into a solid orb for stable arrangement; *Melt* reopens it as liquid for further exploration. *Exudation* develops attached child material while retaining the parent as an organizing reference. Solid objects remain available for grabbing, inspection, and arrangement.

Users bring two bubbles together for fusion and pull a bubble apart with both hands for fission. Closing the hand around a liquid bubble freezes it; sweeping a hand across a solid orb melts it back into liquid. To exude, users pull a small child bubble outward from a larger one, elaborating a subpoint while keeping the parent in place.

The state transition preserves an identifiable working reference, rather than restoring a previous version by chronological undo. Freezing marks material as settled for now; melting makes it available for further exploration. Neither state alone establishes that the content satisfies the task. P01’s recorded revision below illustrates this distinction: reopening precedes the correction of a walking limit.

The vocabulary supports different organizations of a task: factors can be gathered around a decision, or a proposal can be expanded into local details. Figure 4 illustrates these uses with night-market and cafe-event examples. They demonstrate the intended design vocabulary, rather than domain-general performance.

4.3 Content, Context, and Provenance

Each object participates in three related representations. *Content* comprises the current label, draft, or structured facts. *Context* supplies a transformation with the task, presented constraints, current content, and explicit user input. *Provenance* links objects to source objects, requests, and prior transformations. A surviving identifier provides a continuing reference even when its text changes; source links make derivation traceable without guaranteeing semantic preservation.

Spatial grouping acts as an organizational cue. The recorded request configuration establishes explicit task and content context, but no independent proximity-to-prompt selection rule. Users express corrections through typed or

Five Interactive Gestures

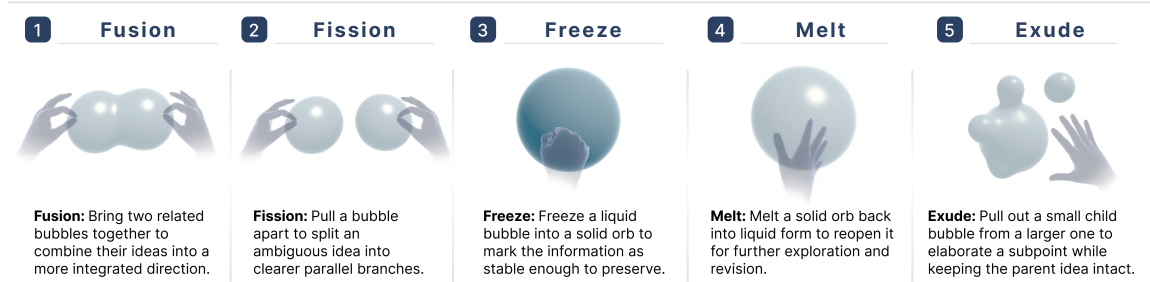


Fig. 3. Liko's five interaction gestures. Users bring bubbles together for fusion, pull a bubble apart for fission, close the hand to freeze, sweep a hand across a solid orb to melt, and pull a child bubble outward to exude. Freezing stabilizes material for organization; melting reopens it for exploration and revision.

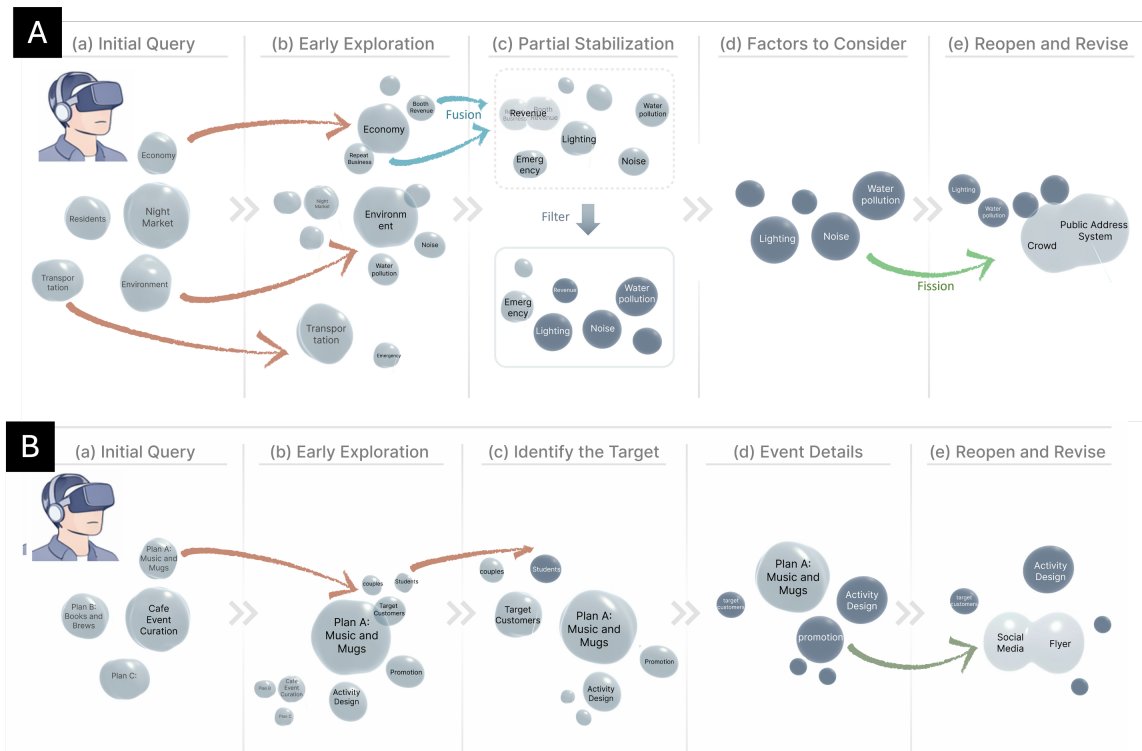


Fig. 4. Illustrative exploration workflows. (A) Night-market planning develops and combines factors, then reopens noise into crowd and public-address concerns. (B) Cafe-event curation develops a proposal and later differentiates promotion into specific channels. These design illustrations are not captured evaluation trials; cafe-event curation differs from the employee-event task.

spoken submissions; the records do not establish implicit selection of a multi-object dependency scope from speech alone. Both Liko and the VR baseline expose drafts through grabbing. Stored histories support analysis, but a dedicated chronological version-comparison or provenance browser is not established by the records.

4.4 A Recorded Revision Sequence

P01’s travel task illustrates how reopening fits into revision. When the daily walking limit tightened from 6 to 3 km, P01 requested a reorganized route through speech and melted the previously frozen walking-limit object. Reopening kept the same object available for further work, but did not immediately correct its content: the next recorded snapshot still showed 6 km. A later update changed the limit to 3 km, and further edits produced a state passing all critical task criteria. The model request was triggered by the spoken input, not by melting. This case separates reopening an object, correcting its content, and achieving task success. Appendix D.10 provides the detailed event sequence; the account is reconstructed from logs rather than recorded views or participant testimony.

4.5 Implementation and Model Configuration

Liqo is implemented in Unity and deployed on a PICO 4 Ultra headset. Its front end maintains the 3D workspace, tracks object states, and handles fusion, fission, freezing, melting, and spatial arrangement. Unity’s XR Interaction Toolkit supports direct grabbing, hovering, and two-handed coordination. These interactions connect the visible liquid and solid objects to persistent information objects and LLM-mediated content transformations.

Liquid objects are rendered as metaballs rather than independent meshes. Nearby liquid nodes form connected clusters, each enclosed by a bounded proxy volume. Within each volume, a signed-distance field combines sphere primitives and short motion trails through smooth unions, allowing adjacent units to blend and separate continuously. A custom Universal Render Pipeline volume shader evaluates the field, estimates surface normals, and applies depth-aware shading to produce a soft, semi-transparent appearance. TextMeshPro glyph layouts embed labels within the liquid form and remain synchronized with the runtime metaball representation.

The backend maintains each object’s content, material phase, parent-child relations, and derivation metadata. Fusion creates an object linked to its sources; fission creates child objects associated with the parent. Freezing and melting switch between liquid and solid visual profiles while retaining the underlying information object. Synchronizing these profiles with persistent state maintains the connection between visible objects and their transformation history. Object identifiers and request links connect successive states for the evaluation’s analysis of material transitions and content changes.

Evaluation requests identify gpt-5.4 for Liqo and the VR baseline, and qwen-plus for Chat and Canvas. Generation settings are temperature 0.2, top- p 1.0, a 4096-token output limit, streaming, and request seed 20260912; an immutable model snapshot was not recorded. The immersive conditions share a base prompt emphasizing distinct branches, revisable drafts, hierarchical context, and preserved constraints. Task-specific instructions are appended, but a travel-oriented profile remains in the shared base across domains. Desktop conditions use a shorter prompt. Appendix D documents the prompts and request outcomes.

5 Evaluation

We evaluated Liqo through planning tasks in which requirements change after participants have begun organizing a solution. The four-condition, within-participant study comprises 13 participants and three tasks per interface (156 trials). The central question is whether the tested configuration supports whole-plan revision while maintaining applicable

Table 2. Implemented and recorded properties of the two immersive configurations. Formal operations are separated from standardized post-task controls.

Property	Liqo	VR baseline
Formal operation vocabulary	Fusion, fission, exudation, freeze, melt.	Fusion, fission, exudation.
Post-task controls	Fuse, split, exude, freeze, melt.	Group, ungroup, duplicate/extract, lock, unlock.
State and layout	Liquid/solid forms; near, staging, and output regions.	Editable spatial objects; recorded positions and access targets.
Intent and generation	Typed/speech input; label and draft; task, constraints, current content, and user edits in context.	Same input modes, output fields, and XR base prompt; task-version suffixes vary.
Stored history	Content states, request links, source-object links, and object phases.	Content states, request links, and source-object links.
Text access	Grab object to read draft; 848 logged readable accesses and 39 post-task reading panels.	Grab object to read draft; 945 logged readable accesses and 39 post-task reading panels.

work within its structured representation. Revision outcomes address this question directly; behavioral measures, artifact ratings, access tests, and self-reports provide complementary evidence.

RQ1: Revision. How successfully and efficiently can users meet changed requirements while maintaining a complete proposal and same-key content continuity?

RQ2: Interaction. How do users organize and reopen material, and how predictable and destructive are the tested operations?

RQ3: Access and cost. How well can users revisit and read material, and what interaction costs do they report?

5.1 Study Design

5.1.1 Conditions and Comparison Logic. Chat provides sequential prompting and responses with a text-editing workspace (Appendix Figure B.2). Its logged content operations include combining sections, splitting outlines, and extracting subsections. Canvas provides persistent cards with spatial arrangement and editing, including merging, splitting, and local extraction. It reuses the formative probe implementation (Figure B.1), rather than replicating Sensecape [31]. VR baseline provides editable spatial objects, fusion, fission, and exudation, but no liquid–solid phase transitions. Liqo adds freezing and melting with the state-dependent representation and workspace described in Section 4.

Liqo–VR is the focal comparison because both use the same model identifier, provider, generation settings, and base system prompt. It evaluates the phase-transition *design package*, including its material representation, available controls, and spatial staging. Chat and Canvas use a different model and shorter prompt (Appendix D); desktop–VR comparisons are descriptive. Model, display, input, and operation differences prevent attributing these cross-family differences to an interface feature.

Table 2 separates formal-task operations from post-task probe controls. The VR baseline has no lock/unlock operation during formal planning; those controls appear only in the standardized post-task probe, where they test expectations about object count and editability. In both immersive conditions, participants open an object’s readable draft by grabbing it. The comparison therefore concerns the complete tested configurations rather than a feature-matched estimate of the liquid–solid encoding alone.

5.1.2 Participants and Data Coverage. We recruited 13 participants (5 female, 8 male; mean age 22.8 years, SD 4.1) from the local university community through departmental mailing lists and word of mouth. Participants used LLMs frequently and represented backgrounds including science, engineering, economics, AI, media arts, and HCI. Appendix Table C.2 provides individual background descriptors. Three had no prior VR experience, seven had limited or moderate experience, one had frequent experience, and two had extensive experience. Participants had no prior exposure to Liqo or the formative probe. The evaluation received approval from the researchers' institutional review board.

All 13 participant identifiers (P01–P13) have one included, non-practice record for every condition–task combination: 39 trials per condition. None of the supplied records is flagged for exclusion. All outcomes tested below have 13 complete participant pairs after averaging the three tasks.

5.1.3 Tasks and Order. Each interface was used for three bounded planning tasks in Chinese. Each task has four versions with different names and budgets; each participant encountered all four versions of each task across conditions. All versions introduced a 20% budget reduction at minute 10, alongside task-specific changes:

- *T1: Travel planning.* Plan a three-day, two-night trip for two people, comparing travel styles and accommodation areas and retaining an alternative route. The change also moves the second afternoon indoors because of rain and reduces the daily walking limit from 6 to 3 km.
- *T2: Employee event.* Coordinate a one-day event with venue, schedule, catering, and team activities. Attendance rises from 40 to 50, including 10 vegetarian meals; the revised venue must be indoors and accessible.
- *T3: Waterfront night market.* Compare permanent construction, a limited trial, and relocation; recommend a decision addressing residents, vendors, visitors, and municipal stakeholders, with exit conditions. The change mandates a small-scale trial and adds a 55 dB estimated-noise ceiling, closing by 21:00, and a 4 m fire-access lane. Noise is a requirement on the proposal, not an acoustic measurement or verified prediction.

Condition order follows four balanced-Latin-square sequences, replicated 4, 3, 3, and 3 times (Appendix D). Task order also varies. The unequal replication at $N = 13$ is not perfect counterbalancing. Different task versions reduce exact repetition without establishing equal difficulty or eliminating learning.

5.1.4 Procedure and Recorded Sources. Participants provided informed consent and completed a demographic questionnaire. Before using each interface, they received a short tutorial and completed a brief practice task. Duration and per-control exposure are not preserved, limiting comparisons of how readily individual controls were learned.

Timestamps place each participant's four condition blocks on four separate days, with one condition and three tasks per day. Within a day, task starts are 40 minutes apart. Each formal task has two phases: participants develop a proposal under the initial requirements for 0–600 s, then revise it after the constraint change for 600–1200 s. Thus formal work totals 240 minutes per participant, excluding post-task probes, gaps, and preparation. There are no recorded pauses within formal tasks and no overlapping trial records.

After each task, a prompt at 1210 s asks the participant to locate a specified earlier item and continue viewing it; the deadline is 1250 s. A standardized operation-expectation probe begins at 1255 s: participants select their expected object count and editability before executing each probed operation. A 143-character Chinese reading passage with ten factual questions follows, then questionnaires. These probes fall outside formal revision time. The reading task uses matched immersive display parameters but measures short-text access rather than comprehension of the participant's own plan; Appendix D.8 gives allocation and presentation details.

Table 3. Measure-to-evidence map. Formal outcomes and post-task probes address different questions; none directly measures psychological commitment.

Question	Measures	Recorded evidence and interpretation
RQ1: Revision	Final success; burst/execution decomposition; restricted revision time; same-key continuity	Critical judgments over content states; within-key semantic correspondence and final-node traceability
RQ1: Artifact	Coherence, coverage, constraint satisfaction, actionability	Independent, condition-blind ratings of complete final artifacts, not labels alone
RQ2: Interaction	Depth; branches; deletion, recreation, rechecks; nondestructive revision	Artifact links, branch lifetimes, and coded actions/segments during formal tasks
RQ2: Probe-state prediction	Object-count/editability match	Recorded responses before standardized post-task operations
RQ3: Access	Revisit success/time; reading accuracy/time	Target-linked readable access; ten factual question responses
RQ3: Cost	Raw NASA-TLX; interface comfort; readability; physical discomfort	Post-task responses; physical demand also reported separately

Records contain timestamped events, explicit inputs, model exchanges, content states, final artifacts, judgments, and questionnaires. State replay is available for all trials; audio and video are not. Logged submissions provide content evidence rather than interview testimony. Each post-change content state follows a semantic-edit event. Ligo has a mean 13.77 such states per trial (range 8–33), and VR has 14.90 (10–28); timing therefore concerns the first *recorded* passing state. Spatial outcomes use target-linked readable-access events. Appendix D documents tracking checks and item-level questionnaire records.

5.2 Measures

Table 3 connects the research questions to their evidence. Appendix D supplies critical constraints and assessment details.

Revision outcomes (RQ1). *Final success* requires every task-specific critical constraint to pass at the formal endpoint. Criteria include changed, newly quantified, or newly mandatory *burst-response criteria* and an unchanged *execution-coverage* criterion requiring a complete actionable proposal. Success refers to this conjunction, not every task requirement or externally verified feasibility. The descriptive decomposition distinguishes responding to changes from maintaining or completing the underlying plan, although some burst-response judgments also include existing completeness requirements (Appendix D). *Restricted revision time* is elapsed time from the change to the first recorded state passing all critical constraints, capped at 600 s. Non-attaining trials receive the observation limit rather than being discarded; 600 s is not their actual completion time. Later edits can invalidate an earlier solution, so first attainment and final success are separate outcomes.

Same-key continuity is the fraction of pre-change atomic units judged still applicable that have a semantic match among final candidates with the same canonical key and a traceable final node. An atomic unit is one clause or value span within a structured fact, so unchanged activities can remain eligible even when another activity in the same itinerary changes. We repeat the calculation with whole structured facts as a coarse-grain sensitivity analysis. Both rules exclude obsolete information. Because RET7 candidate packets were restricted to the same key, this measure assesses continuity within the implemented structured representation; it cannot detect a paraphrase or migration stored under a different key and is not a whole-artifact semantic-retention rate.

Organization, cost, and predictability (RQ2). Hierarchy depth is computed from semantic parent–child links in final artifacts. Active branches are the time integral of branch lifetimes divided by formal task duration. Post-change cost counts use common semantic labels for deleted units, recreated units, and repeated checks, not a mixture of character deletions and object manipulations. Nondestructive revision is the proportion of valid revision segments retaining traced facts without delete/recreate actions. The supplied segmentation rule groups edits with the same intent and gaps of at most 30 s; validity requires visible constraint improvement or an added explicit rationale. These are coding-dependent behavioral measures, not direct observations of cognition.

Operation-state prediction uses only standardized post-task probes with a recorded pre-operation response: a match requires the selected object count and editability to equal the observed state. Native formal-operation logs also contain an expected state, but this is a specified target, not a participant prediction. Because native controls and documented learning opportunities differ, this outcome describes prediction for the tested probes rather than the intuitiveness of the material metaphor. Freeze/melt counts are interface-use descriptions rather than cross-interface measures of commitment.

Artifact quality. Two independent raters, blind to condition, evaluated complete final artifacts for coherence, coverage, constraint satisfaction, and actionability on 1–7 scales. They assessed the plans’ content, not the brevity of interface labels. Shared anchors apply across dimensions (Appendix D). We average the two ratings per dimension and report the four-dimension mean descriptively. Agreement uses two-way absolute-agreement average-measure ICC(A,2), with participant-cluster bootstrap intervals.

Revisitation, reading, and experience (RQ3). Revisit success requires readable access to the specified target within 40 s; restricted revisit time is the first such access latency capped at 40 s for unsuccessful trials. Reading accuracy uses exact answer matches across ten questions, and response time averages question-to-answer intervals across all ten, including incorrect answers. Workload is unweighted Raw NASA-TLX: the mean of all six 0–100 items, with the performance response reversed into failure so higher values indicate greater demand or worse experience. Interface comfort asks how comfortable the interface was throughout the task, from 1 (very uncomfortable) to 7 (very comfortable). Readability asks how easy the task text and content were to read, from 1 (very difficult) to 7 (very easy); physical discomfort ranges from 1 (none) to 7 (very strong). We also report physical demand separately. These brief assessments do not establish long-term fatigue or accessibility.

5.3 Analysis

This is an exploratory analysis, not a preregistered confirmatory test. We compute trial measures from source-linked records, then average the three tasks within participant and condition. The inferential unit is the participant ($n = 13$), not the 39 trials per interface. Two-sided paired sign-flip tests enumerate all $2^{13} = 8192$ sign assignments for Liqo–VR differences. These rely on sign exchangeability under the null; they are not design-based randomization tests of the full four-period assignment. Mean differences have 95% percentile intervals from 20,000 paired participant-bootstrap samples.

All main-text p_A values use Holm adjustment across the complete set of 23 contrasts. Appendix Table D.7 additionally reports adjustment within five exploratory reporting families: revision outcomes (3), artifact dimensions (4), process/predictability (7), revisit/reading (4), and experience (5). These families are not preregistered. Bootstrap intervals are unadjusted, not simultaneous, and need not agree with adjusted tests, especially for discrete small-sample outcomes. Desktop results remain descriptive; nonsignificance is not evidence of equivalence. A descriptive sensitivity analysis

Table 4. Outcomes by condition: mean $\pm$ SD of 13 participant means, each averaging three tasks. Percentages use equal task and participant weights. Desktop-VR differences are descriptive because model, prompt, display, and input differ.

Measure	Chat	Canvas	VR baseline	Liqo
Final success (%)	38.46 $\pm$ 26.69	74.36 $\pm$ 27.74	61.54 $\pm$ 18.49	97.44 $\pm$ 9.25
Restricted revision time (s)	463.72 $\pm$ 75.30	292.23 $\pm$ 94.24	434.86 $\pm$ 47.85	186.57 $\pm$ 55.47
Same-key continuity (%)	91.84 $\pm$ 5.54	94.32 $\pm$ 4.18	91.18 $\pm$ 6.31	99.74 $\pm$ 0.66
Mean artifact rating	5.77 $\pm$ 0.45	6.15 $\pm$ 0.58	5.99 $\pm$ 0.36	6.65 $\pm$ 0.12
Restricted revisit time (s)	24.95 $\pm$ 6.53	22.78 $\pm$ 7.15	24.68 $\pm$ 6.63	6.67 $\pm$ 2.14
Raw NASA-TLX	50.71 $\pm$ 10.11	49.70 $\pm$ 5.67	57.07 $\pm$ 10.42	38.33 $\pm$ 13.59

adjusts task-level Liqo-VR differences for domain and the difference in condition-block position without treating task pairs as independent participants. The burst/execution decomposition adds no significance tests to this family.

Two fluent Chinese readers with research-coding experience independently coded constraints, continuity, and revision segments. They received written rules, a 60-minute task briefing, and 90 minutes of coding instruction, followed by nine practice and 18 calibration cases. Four calibration cases initially differed. After discussing training material and clarifying three boundary rules, both independently rechecked all 27 cases; the codebooks were frozen before formal coding.

Formal packets retained task requirements and structured evidence while excluding explicit condition, participant, session, and order identifiers, native operation names, interface media, prior judgments, and results. Continuity packets supplied only same-key final candidates. Coders received different presentation orders, had no access to each other's files or the condition mapping, and received no directional hypotheses. They separately locked 28,402 targets before merging and statistical comparison: 20,630 state-criterion judgments, 2,935 whole-fact and 4,018 atomic continuity units, and 819 revision segments. Every paired label agreed, so the appointed adjudicator resolved no formal disagreement. This documents consistent application of the codebooks, not external validation: repeated states are dependent, and content patterns may still permit condition inference. Appendix D provides the protocols, masking rules, and completion chronology.

The supplement retains source-linked judgments and verification outputs. These checks and the descriptive decompositions add no tests to the 23-contrast family. Artifact ratings use a separate assessment process.

5.4 Results

Table 4 summarizes all four conditions. Figures 5 and 6 summarize participant-level outcomes for the focal Liqo-VR comparison.

5.4.1 Successful Revision and Structured Continuity (RQ1). Final success occurred in 38/39 Liqo trials (97.4%) and 24/39 VR-baseline trials (61.5%), a paired difference of 35.9 percentage points (95% CI [25.6, 46.2], $p_A = .0107$). Chat and Canvas had 15/39 (38.5%) and 29/39 (74.4%) successful final artifacts. Desktop rates contextualize performance but cannot identify an interface-only effect.

The larger endpoint difference concerned the completeness of the proposal. Burst-response criteria passed in 38/39 Liqo and 33/39 VR trials; execution coverage passed in 39/39 and 26/39. Table 5 separates the components and the execution transitions. Although fewer Liqo proposals had complete execution coverage at the change point, all 19 initially incomplete proposals passed that criterion at the endpoint, as did all 20 initially complete proposals. VR

Table 5. Descriptive components of the focal revision comparison, 39 trials per condition. Execution transitions compare the 600-s assessment with the endpoint; they do not describe every intermediate state.

Outcome	VR baseline	Liqo
Execution coverage at change	27/39	20/39
Burst-response criteria at endpoint	33/39	38/39
Execution coverage at endpoint	26/39	39/39
All critical criteria at endpoint	24/39	38/39
Execution: pass $\rightarrow$ pass	23/39	20/39
Execution: pass $\rightarrow$ fail	4/39	0/39
Execution: fail $\rightarrow$ pass	3/39	19/39
Execution: fail $\rightarrow$ fail	9/39	0/39

combined three repairs with four regressions. These paired endpoints describe maintenance and completion across the revision interval, without establishing uninterrupted coverage between observations.

Restricted time to burst-response attainment averaged 159.1 s for Liqo and 402.5 s for VR; time to execution coverage averaged 85.3 and 137.3 s. Thus the overall outcome combines satisfaction of changed requirements with maintaining or completing an executable proposal. The decomposition is descriptive and does not identify separate causal mechanisms. Appendix Tables D.1 and D.2 provide the full criterion mapping and all four conditions.

Liqo’s mean restricted revision time was 186.6 s, compared with 434.9 s for VR: difference -248.3 s (95% CI $[-281.8, -211.6]$, $p_A = .0056$). One Liqo and 12 VR trials never attained a passing state within the window; corresponding counts were 18 for Chat and 3 for Canvas. Canvas therefore attained a passing state in 36 trials but ended with one in only 29. First attainment cannot substitute for final success.

Atomic-unit same-key continuity averaged 99.74% in Liqo and 91.18% in VR, an 8.56-point difference (95% CI $[5.42, 12.13]$, $p_A = .0056$). The underlying counts are 658/660 and 602/657 eligible units, respectively; the reported means average trial ratios within participants rather than pooling units. Within the issued same-key candidate sets, every eligible unit’s semantic-match and traceability labels coincide, so the semantic, traceability, and joint rates are numerically identical. With whole structured facts as the unit, the corresponding means are 99.60% and 89.63%, a descriptive 9.97-point difference in the same direction. These results support continuity under stable structured keys, not exhaustive semantic retention across the final artifact. Appendix D.5 gives both denominators and content examples.

Adjustment for task and block position yields similar descriptive differences: 35.7 percentage points in final success, -247.7 s in restricted revision time, and 8.6 points in atomic-unit same-key continuity. Task-specific success differences are heterogeneous: 7.7 points for travel, 61.5 for events, and 38.5 for the market task. VR’s two unsuccessful travel trials both fail execution coverage, with one also failing the walking limit. Seven of its eight unsuccessful event trials and four of its five unsuccessful market trials fail execution coverage. Thus much of the domain variation involves maintaining a complete executable proposal, rather than merely meeting the new numerical ceilings. Liqo’s single final failure is P05’s travel plan, costing CNY 4,646 against a CNY 4,160 budget. Appendix D.6 lists overlapping failure categories; the small task subsets do not establish domain generality.

5.4.2 Organization, Predictability, and Revision Cost (RQ2). The formal Liqo logs contain 497 freeze and 449 melt attempts across 39 tasks. Of these, 473 freezes and 426 melts reach their specified target states. Across all formal Liqo operation types, 71 attempts miss their specified target and 31 have a recorded successful recovery. The other 40 are right-censored at task end: 39 have no linked recovery attempt, and one has an attempt without a recorded successful

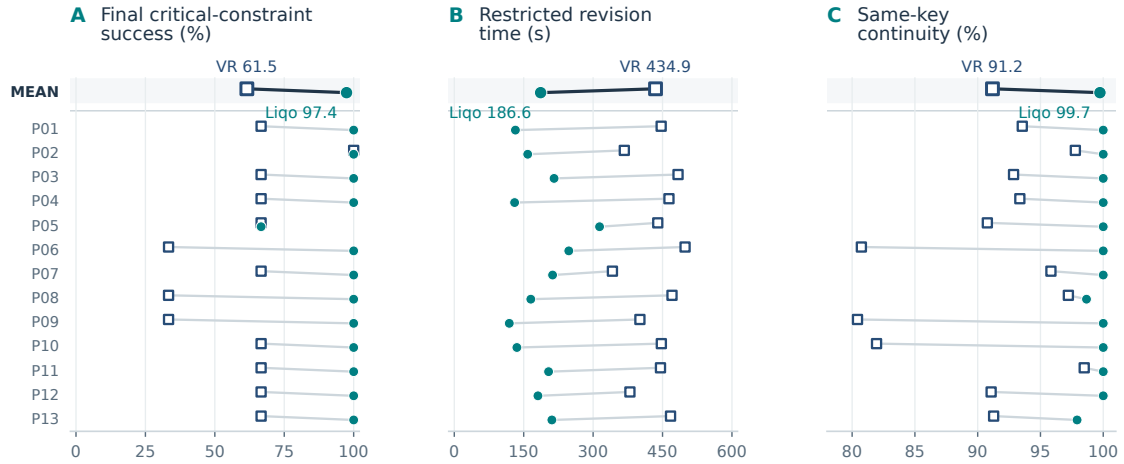


Fig. 5. Focal Ligo–VR revision outcomes. Each horizontal connector joins one participant’s condition means over three tasks; the shaded top row shows condition means. Participant rows align across panels. Paired differences, bootstrap intervals, and Holm-adjusted tests appear in the text and Appendix Table D.7. Restricted time includes non-attainment at 600 s; continuity uses eligible atomic units under the same key and its axis is restricted to 78–101% for legibility. Table 4 gives desktop context.

recovery. They are not 40 observations of data loss or abandonment. In P12’s event task, for example, a failed exudation leaves the parent and its two fact keys in place; failure to create the target children differs from erasing their content (Appendix D.6).

Post-change deletions averaged 0.21 semantic units in Ligo versus 1.85 in VR (difference -1.64 , 95% CI $[-2.00, -1.28]$, $p_A = .0056$); recreation averaged 0.15 versus 0.79 (difference -0.64 , CI $[-1.00, -0.31]$, $p_A = .0293$). Nondestructive valid revisions averaged 94.7% versus 64.0% (difference 30.7 points, CI $[18.4, 42.4]$, $p_A = .0117$). These describe less destructive editing alongside better task outcomes, rather than interpreting fewer actions alone as better performance.

Final semantic depth was 4.15 in Ligo and 3.97 in VR ($p_A = .2813$). Active branches averaged 1.34 and 1.59 ($p_A = .2227$); repeated checks averaged 6.95 and 8.28 ($p_A = .2813$). These comparisons do not establish deeper organization, more parallel exploration, or fewer rechecks. Post-task state-prediction matching averaged 94.8% in Ligo and 79.3% in VR (difference 15.4 points, CI $[8.9, 22.6]$, $p_A = .0063$). Restricting descriptively to fusion, fission, and exudation yields 95.3% versus 80.0%. Because native controls, frequencies, and per-control learning opportunities are not matched, these outcomes characterize the tested probes rather than demonstrating that the material metaphor is intrinsically more intuitive. Appendix D.7 provides counts and mappings.

5.4.3 Artifact Assessment. Ligo’s mean artifact rating was 6.65, compared with 5.99 for VR, 6.15 for Canvas, and 5.77 for Chat. Ligo–VR dimension differences were 0.59 for coherence, 0.63 for coverage, 0.45 for constraint satisfaction, and 0.97 for actionability (all $p_A < .021$; Appendix Table D.7). ICC(A,2) was .889 for coherence (95% CI $[\.837, \.921]$), .883 for coverage ($[\.775, \.936]$), .921 for constraint satisfaction ($[\.795, \.961]$), and .937 for actionability ($[\.919, \.951]$). These estimates quantify agreement between assessments, not external verification of plan quality. Several scores approach the upper scale bound, limiting sensitivity to differences among highly rated artifacts.

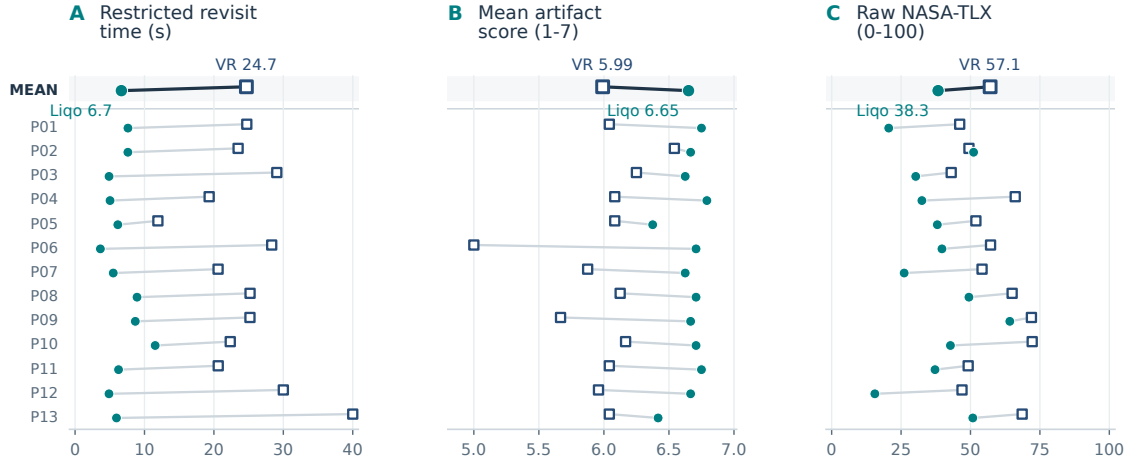


Fig. 6. Participant-level access, artifact quality, and workload in the focal comparison. Each horizontal connector joins one participant’s VR-baseline and Liqo means over three tasks; the shaded top row shows condition means. Participant rows align across panels. Revisit time includes unsuccessful trials at the 40 s cap. Artifact ratings average two raters across four dimensions; that panel’s axis is restricted to 4.8–7.0 for legibility. Raw NASA-TLX is the unweighted mean of six 0–100 items.

5.4.4 Revisitation, Reading, and Experience (RQ3). All 39 Liqo trials met the post-task revisit criterion, compared with 23/39 VR trials. Restricted revisit time averaged 6.67 s versus 24.68 s (difference -18.01 s, CI $[-21.95, -14.31]$, $p_A = .0056$). This tests access under the recorded display arrangements; it does not isolate spatial memory from target distance, occlusion, or the workspace scaffold.

The secondary reading probe also favored Liqo on accuracy and response time (both $p_A = .0056$; Appendix D.8). These results characterize the tested short-text panels; passage difficulty was not independently equated.

Raw NASA-TLX averaged 38.33 for Liqo and 57.07 for VR (difference -18.74 , CI $[-24.10, -13.25]$, $p_A = .0063$). Interface comfort and readability also favored Liqo (differences 1.49 and 1.13; both $p_A = .0056$), and physical demand was lower by 18.85 points ($p_A = .0056$). Physical discomfort did not differ reliably (1.79 versus 1.92; difference -0.13 , CI $[-0.38, 0.18]$, $p_A = .5176$). These self-reports complement the task outcomes without establishing long-term fatigue or accessibility (Figure 6).

6 Discussion

6.1 Revision Includes Maintaining a Usable Proposal

The Liqo–VR comparison favors the tested Liqo configuration in whole-plan revision, same-key continuity, and target access. The decomposed results clarify what the task advantage involves. The endpoint difference in complete execution coverage is larger than the difference in satisfying burst-response criteria. Liqo’s initially incomplete proposals all reach execution coverage by the endpoint, while VR combines repairs and regressions. These outcomes connect the design problem in the Introduction to the practical work of revision: accommodating a new requirement can require completing or maintaining the surrounding proposal as well as changing the affected value.

This finding gives a specific reason to evaluate more than the changed constraint. An event plan may accommodate additional attendees while omitting execution support; a travel plan may adopt a walking limit without retaining a complete schedule. A measure focused only on the new requirement would count these partial outcomes differently from the whole-plan criterion. Conversely, initial completeness differs between the conditions, so the observed advantage cannot be described solely as preserving already-finished plans. The comparison concerns the complete revision episode, and the descriptive decomposition identifies its components without attributing them to separate interface mechanisms.

6.2 Reopening Has More Than One Successful Outcome

The cases reveal consequences that aggregate task performance alone does not capture. In P01's travel task, the walking-limit object returns to liquid with its identity intact, but its content remains at 6 km before a later update changes it to 3 km. Reopening supplies a target for continued work; explicit intent and content updates address the changed requirement. P08's final plan passes the critical criteria despite lacking same-key continuity for an accommodation alternative. P05 retains a structured plan that exceeds the revised budget. P12's failed exudation preserves its parent while failing to create the requested children.

These cases distinguish three achievements: *restoring manipulability*, *maintaining useful content and lineage*, and *passing task criteria*. Their separation changes what an evaluation must inspect. Counting successful object transitions would miss incorrect or incomplete content. Counting surviving nodes would miss semantic changes. Counting only final task success would miss discontinuity for alternatives still useful for comparison. Branching, pinned-card, and versioned LLM workspaces can face the same distinctions, even without material imagery.

6.3 Making the Material Contract Inspectable

AI metaphors foreground some relations while concealing others [22, 23]. Liko makes this issue concrete at the boundary between an object transformation and a content update. A phase animation communicates working status, but it cannot by itself show whether a walking limit has changed. P01's sequence demonstrates that reopening and content correction occur at different recorded states. A material contract therefore needs to explain what the state controls, what remains in progress, and what has actually changed.

Distinguish working status, generation progress, and task checks. P01's sequence motivates separate indications of *ready to edit*, *generation in progress*, and *content updated*. P05's budget failure further motivates presenting task-level checks independently of a solid appearance. These are design implications of the observed separations, not features evaluated in Liko. They extend the state vocabulary by making consequential content changes inspectable without requiring the material representation to certify correctness.

Expose the content affected by a changed requirement. P01's second-day itinerary contains unchanged morning, lunch, and evening activities alongside a superseded outdoor afternoon. A visually coherent object can therefore contain parts with different revision needs. Interfaces could expose the affected clauses while retaining the larger object as an organizing reference. Detailed inspection should accompany reopening so users can check numerical limits and alternatives beyond compact labels. The present revisit and short-text probes establish bounded access, leaving sustained draft comparison less well characterized.

Show recoverable material after an incomplete transformation. In P12's failed exudation, the parent remains available although the children are not created. Showing what remains usable would support recovery more directly than a

generic failure indication. Evaluations should likewise distinguish structural failure, semantic loss, and an unresolved recovery attempt. The 40 operation failures without a recorded successful recovery cannot establish 40 cases of deletion or abandonment.

For evaluation, these implications require a clear unit of continuity and candidate-search scope. Atomic coding retains P01's unchanged itinerary clauses, whereas a whole-fact rule excludes the compound itinerary; both produce the same direction of Liqo-VR difference. Neither rule detects semantic migration across keys because the issued candidates are restricted to the same key. A whole-artifact follow-up should search for semantic survival independently of key continuity and assess traceability separately. This would distinguish losing useful content from reorganizing it under a new representation.

6.4 The Role of the Immersive Configuration

VR combines bodily manipulation, spatial staging, and object-level state control. Better target access and lower reported workload apply to that configuration, including its visibility, distance, occlusion, and controls. Physical discomfort did not differ reliably; the study establishes no general physical benefit of immersion.

The baseline offers editable objects and readable drafts but lacks formal-task lock/unlock controls. Matching state controls, layout, content, and history access would test the added value of material encoding. A matched desktop realization would address a separate question: the role of immersion.

6.5 Limitations and Scope

Participants and tasks. The formative input comes from ten STEM students and one travel scenario, with incompletely documented analysis. The evaluation has 13 participants from one university community and bounded individual planning tasks. Repeated domains, task versions, unequal order replication, four visits, and differing initial completeness limit interpretation. Participant aggregation and descriptive task/order adjustment do not eliminate learning, novelty, fatigue, or carryover.

Comparison and inference. The immersive comparison evaluates the whole design configuration. Desktop comparisons additionally confound model and prompt, immutable model snapshots are unavailable, and the shared XR prompt retains a travel profile in other domains. All tests are exploratory. Holm adjustment covers 23 paired tests, while bootstrap intervals are unadjusted. The burst/execution decomposition is post hoc and descriptive. Small task subsets do not establish domain generality.

Measurement. Coding agreement does not externally validate judgments: states are dependent and content patterns may reveal condition. Same-key continuity omits cross-key migration; artifact ratings may share bias or encounter ceilings. Operation probes differ in controls and documented learning exposure; reading variants lack difficulty calibration. Brief self-reports establish neither long-term experience nor simulator sickness. Written-plan criteria do not externally verify costs, accessibility, noise, or safety.

Implementation and records. Logs support state reconstruction, but comparative-study audio, video, and screenshots are unavailable for examining participant reasoning or rendered views. Stored derivation links do not establish a dedicated history-comparison interface. These boundaries limit claims about experienced mechanisms.

7 Conclusion

Liqo gives provisional choices an explicit, reversible working state through liquid–solid interaction in VR. In the tested planning tasks, it supported more successful whole-plan revision than the VR baseline, particularly in maintaining or completing an executable proposal. Yet reopening could precede content correction, and critical criteria could pass while alternatives lost structured continuity. Reopenable LLM workspaces should therefore make working status, content changes, and task checks separately inspectable, so users can build on selected material while retaining a clear route back into revision.

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A Canvas Probe Implementation

The formative probe was implemented in HTML, CSS, and JavaScript. Its presentation layer combines a chat panel, spatial canvas, floating controls, and a saved-card area. The interaction layer maintains cards, links, groups, and session state. An OpenAI-compatible integration layer handles streaming generation, request locking, and error handling; the deployment uses Qwen-Plus through DashScope.

Cards hold generated or user-written text and support moving, editing, deleting, splitting, and prioritizing. Users can compare selected cards, produce a comparison summary, group and rename cards, collapse or expand groups, and add directed SVG links. Local follow-up generation elaborates selected content; a global conclusion operation integrates visible material. Saving a card removes it from the active canvas while keeping it accessible for restoration. These affordances supplied the design setting in Section 3; they were not intended as a feature-for-feature reproduction of another published system.

The prototype records operation types, timestamps, object identifiers, and action parameters, including card manipulation, comparisons, grouping, generation, saving, and restoration. Session export includes message history, cards, links, groups, and events. These traces describe workspace activity; they do not by themselves identify a user's cognitive phase. Figure B.1 shows the interface, and Figure B.2 shows the separate sequential interface used in the evaluation.

B Interface Illustrations

Figures B.1 and B.2 document the desktop interfaces. Lipo's overview, material operations, and illustrative workflows appear in Figures 1–4 in the main text.

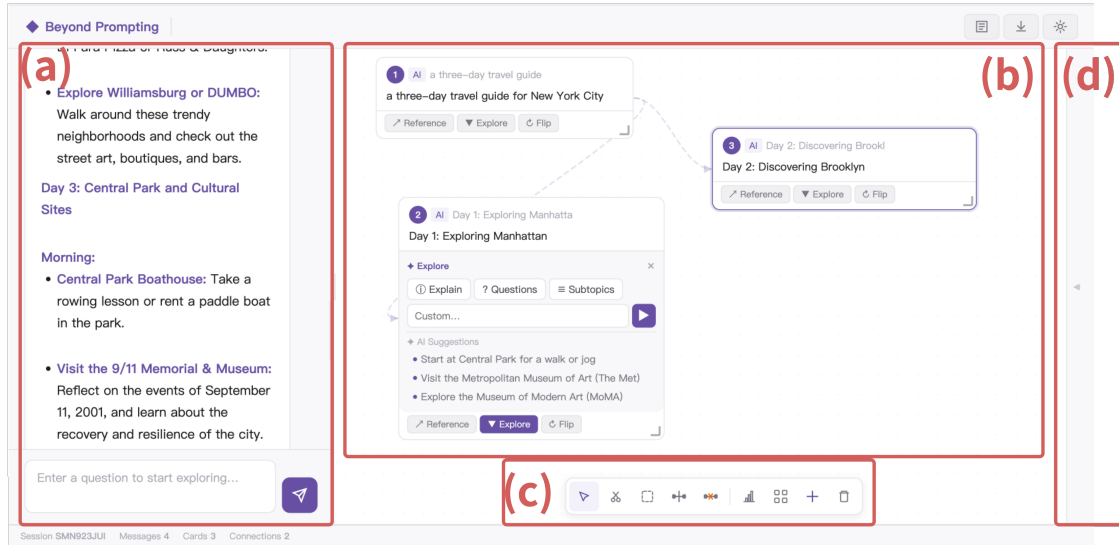


Fig. B.1. Canvas probe and evaluation interface. Persistent cards support local exploration, spatial arrangement, grouping, comparison, and saved material. The probe provides design input rather than a replication of Sensecape.

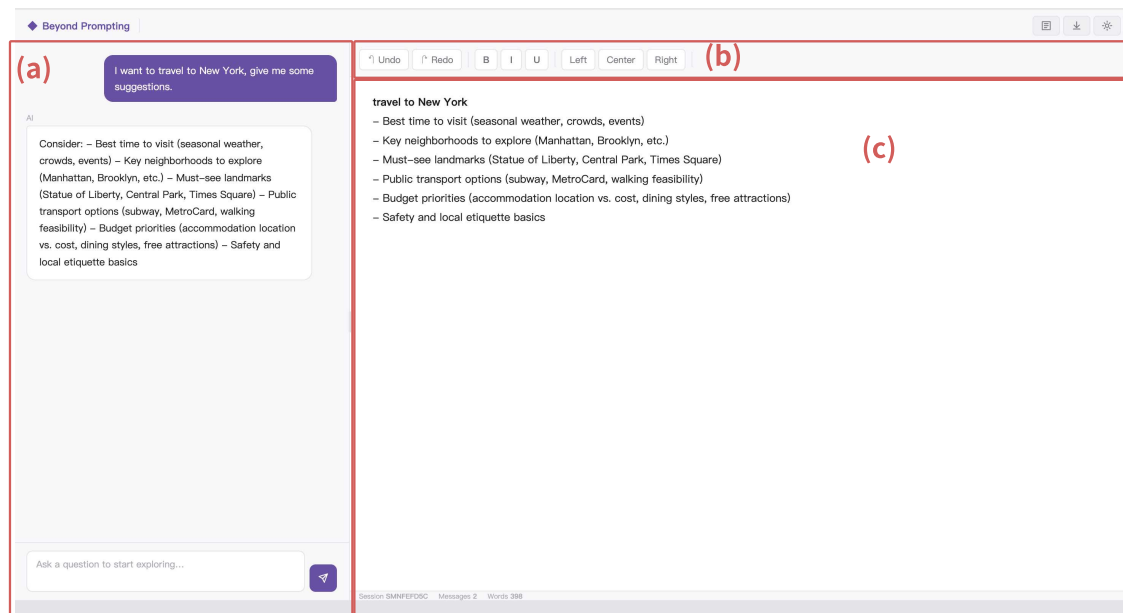


Fig. B.2. Chat evaluation interface. The conversation panel presents sequential responses; an action bar and rich-text workspace support composing prompts and revising generated content.

C Participant Backgrounds

The samples are distinct; identifiers are local to each study. Tables C.1 and C.2 report self-described backgrounds. UG denotes undergraduate; MSc denotes master’s student. Weekly+ denotes the original several-times-per-week category; Daily groups daily-use responses. Experience categories are descriptive self-reports, not performance measures.

Table C.1. Formative-study participants. Explore denotes frequency of exploratory tasks; Canvas denotes prior spatial/canvas-tool experience.

ID	Gender	LLM use	Explore	Canvas	Background
P01	Female	Weekly+	Often	None	Molecular science/engineering (UG)
P02	Male	Weekly+	Often	Limited	Molecular science/engineering (UG)
P03	Male	Weekly+	Often	Limited	Molecular science/engineering (UG)
P04	Male	Daily	Often	Limited	Molecular science/engineering (UG)
P05	Female	Daily	Very often	Moderate	Big data science (UG)
P06	Male	Daily	Very often	Moderate	Computer Science (UG)
P07	Male	Daily	Often	Moderate	Computer Science (UG)
P08	Female	Weekly+	Often	Limited	Big data science (UG)
P09	Male	Daily	Often	Moderate	Big data science (UG)
P10	Female	Daily	Very often	Moderate	Computer Science (UG)

Table C.2. Evaluation participants. All 13 are represented in every condition and task. VR experience retains the reported None, Limited, Moderate, Frequent, and Extensive categories.

ID	Gender	LLM use	Main uses	VR	Background
P01	Female	Daily	Study support; brainstorming; creative ideation	Limited	Big data science (UG)
P02	Female	Weekly+	Technical explanation; report writing; travel planning	None	Molecular science/engineering (UG)
P03	Male	Weekly+	Concept clarification; problem solving; summarization	None	Mathematics (UG)
P04	Male	Weekly+	Literature search; information comparison; study support	Limited	Soft matter science/engineering (UG)
P05	Male	Weekly+	Planning; report drafting; information comparison	None	Economics (UG)
P06	Female	Weekly+	Technical explanation; coding and debugging; paper summarization	Moderate	Information/communication engineering (MSc)
P07	Male	Weekly+	Coding; experiment planning; debugging	Limited	Applied AI (MSc)
P08	Female	Weekly+	Creative ideation; writing support; content organization	Limited	Media arts (UG)
P09	Male	Daily	Research synthesis; coding; brainstorming	Moderate	AI (MSc)
P10	Male	Weekly+	Project planning; coding; game-related ideation	Extensive	AI (MSc)
P11	Female	Daily	Study support; coding; brainstorming	Limited	AI (UG)
P12	Male	Daily	Research planning; literature synthesis; paper drafting	Frequent	HCI researcher
P13	Male	Daily	Research support; brainstorming; writing support	Extensive	HCI (MSc)

D Evaluation Details and Full Contrasts

D.1 Order Allocation

The condition sequences were Chat–Canvas–Liqo–VR ($n = 4$), Canvas–VR–Chat–Liqo ($n = 3$), VR–Liqo–Canvas–Chat ($n = 3$), and Liqo–Chat–VR–Canvas ($n = 3$). Each participant completed three tasks in each condition on a separate day. The analysis supplement retains participant-level condition order, within-block task order, and task-version allocation. All four versions of each task occur for each participant, but the unequal sequence replication and version assignment preclude perfect balancing.

D.2 Constraint Criteria and Observation Windows

The final-success criterion is the conjunction of four critical judgments for T1, seven for T2, and six for T3. Budget judgments require all five cost components and a total no greater than the version-specific revised budget. For T1, the other criteria cover the walking limit, an executable second-afternoon indoor alternative, and complete three-day scheduling with accommodation, meals, and transport. For T2, they cover capacity for at least 50, at least 50 meals including 10 vegetarian meals, an indoor venue, accessible entry, and complete schedule, food, activities, and execution support. For T3, they cover estimated noise at most 55, closing time at most 21:00, fire-lane width at least 4 m, a limited-trial decision, and coverage of four stakeholder groups and an exit rule. These thresholds assess written plans, not their externally verified safety or feasibility.

Additional required items are separate from the critical-success conjunction: T1 retains travel-style comparison, alternative accommodation and route, cultural activities, and nightlife; T2 retains a venue alternative; T3 retains an alternative proposal. Thus passing critical constraints is not the same as satisfying every recorded requirement. The source supplement retains the exact Chinese instructions and version-specific budgets.

The first-valid-state analysis considers all recorded critical judgments from the constraint change at 600 s through formal task end at 1200 s. Every trial has its first post-change judgment at exactly 600 s. Every later content state in both immersive conditions is triggered by a semantic-edit event. Liqo records a mean 13.77 states per trial (range 8–33), compared with 14.90 (10–28) in VR. Capping at 600 s defines a restricted outcome including non-attainment, not a mean among finishers.

Table D.1 maps every critical criterion from the initial instruction to the 600-s change and final judgment. T3 initially asks participants to compare permanent construction, a limited trial, and relocation and then reach a decision; the burst changes that open decision into a policy mandate to conduct a small-scale trial. The resulting decision judgment belongs to the burst-response group. This group includes tightened or newly mandatory requirements as well as newly introduced ones. Some judgments combine changed thresholds with existing completeness requirements, such as listing all five budget components; the decomposition retains those whole judgments and does not isolate only their changed subparts. In each task, the separate proposal-completeness criterion is unchanged execution coverage. The “initial success” field in Table D.4 is execution coverage at the 600-s assessment, not success against requirements that have just been presented. The revisit analysis uses the separate 1210–1250 s window. Standardized expectation operations begin after 1255 s and must not be combined with native formal operations. In particular, a native operation’s specified target state is not a recorded participant expectation.

Table D.1. Explicit mapping of every critical criterion into the descriptive decomposition. The mapping compares the issued initial instruction with the 600-s change rather than inferring a group from the criterion identifier.

Task	Criterion	Initial role	Change at 600 s	Group/final criterion
T1	budget	Plan within the initial budget ceiling.	The ceiling is reduced by 20%.	Burst-response: Five cost components; total at or below the revised budget.
T1	walk	Daily walking of at most 6 km is preferred.	An injury lowers the daily limit to 3 km.	Burst-response: Daily walking at most 3 km.
T1	day2_indoor	No indoor requirement is specified for day 2 afternoon.	A weather warning requires an executable indoor alternative.	Burst-response: Executable indoor alternative for day 2 afternoon.
T1	execution	Provide the full three-day schedule, lodging, meals, and transport.	Unchanged; the proposal must remain complete while being revised.	Unchanged execution: Complete three-day schedule, lodging, meals, and transport.
T2	budget	Plan within the initial budget ceiling.	The ceiling is reduced by 20%.	Burst-response: Five cost components; total at or below the revised budget.
T2	capacity	Plan for 40 employees.	Attendance increases to 50.	Burst-response: Capacity of at least 50.
T2	meals	Integrate catering for the original 40-person event.	Recheck catering for 50 attendees.	Burst-response: At least 50 meals.
T2	vegetarian	Consider vegetarian employees without a quantity.	Exactly 10 attendees now require vegetarian meals.	Burst-response: At least 10 vegetarian meals.
T2	indoor	Compare two venues; an indoor venue is not mandatory.	The venue must be indoors.	Burst-response: Indoor venue.
T2	accessible	Consider accessibility.	The indoor venue must be accessible.	Burst-response: Accessible entrance.
T2	execution	Provide schedule, food, activities, and execution support.	Unchanged; the proposal must remain complete while being revised.	Unchanged execution: Complete schedule, food, activities, and execution support.
T3	budget	A trial budget is stated alongside the comparison of alternatives.	The ceiling is reduced by 20%.	Burst-response: Five cost components; total at or below the revised budget.
T3	noise	Discuss resident risks; no numerical noise ceiling is specified.	Estimated design noise must be at most 55 dB.	Burst-response: Estimated noise at most 55 dB.
T3	closing	No closing-time ceiling is specified.	Operations must stop by 21:00.	Burst-response: Closing time at or before 21:00.
T3	lane	No fire-lane width is specified.	A 4 m fire lane becomes mandatory.	Burst-response: Fire-lane width of at least 4 m.

Continued on next page

Table D.1 (continued)

Task	Criterion	Initial role	Change at 600 s	Group/final criterion
T3	decision	Compare permanent construction, limited trial, and relocation, then reach a decision.	Policy now mandates conducting a small-scale trial.	Burst-response: Decision to conduct a limited trial.
T3	execution	Cover four stakeholder groups and specify exit conditions.	Unchanged; these modules remain required in the revised proposal.	Unchanged execution: Four stakeholder groups and an exit rule.

Table D.2. Descriptive decomposition of revision outcomes across 39 trials per condition. Burst-response criteria are mapped individually in Table D.1; execution is the unchanged criterion requiring a complete actionable proposal. Transitions compare execution coverage at the 600-s assessment with the final artifact. First-attainment times are restricted to 600 seconds and average three tasks within each participant before the condition mean.

Outcome	Chat	Canvas	VR baseline	Liqo
Initial execution coverage (600 s)	18/39	25/39	27/39	20/39
Burst-response criteria, final	24/39	38/39	33/39	38/39
Execution coverage, final	25/39	29/39	26/39	39/39
All critical criteria, final	15/39	29/39	24/39	38/39
<i>Execution transition, 600 s $\rightarrow$ final</i>				
Pass $\rightarrow$ pass	12/39	19/39	23/39	20/39
Pass $\rightarrow$ fail	6/39	6/39	4/39	0/39
Fail $\rightarrow$ pass	13/39	10/39	3/39	19/39
Fail $\rightarrow$ fail	8/39	4/39	9/39	0/39
Burst-response time (s)	431.0	266.1	402.5	159.1
Execution coverage time (s)	227.2	106.5	137.3	85.3
All critical criteria time (s)	463.7	292.2	434.9	186.6

D.3 Model Requests, Prompts, and Source Audit

Section 4 reports models and generation settings. Desktop prompts request concrete, revisable label/draft JSON with preserved facts and explicit tradeoffs. XR prompts additionally specify distinct branches, hierarchy-aware refinement, committed constraints, user edits, and rejected directions. Liqo and VR share this base, but task-title suffixes vary with assigned versions, so their complete prompts differ. The XR travel profile remains in event and market requests and may influence domain-specific output in both VR conditions. The analysis supplement preserves the actual shared base prompts and exact task instructions.

Of 1767 requests, 1684 succeed, 81 return errors, and two time out. Chat has 355 requests (14 failures), Canvas 659 (31), VR 398 (15), and Liqo 355 (23). There are no cache hits. All 68 retries retain original prompts and generation settings; not every failure has a retry. These descriptive counts are not independent observations. Trials retain the elapsed cost of errors and retries.

All 156 sessions have unique participant-condition-task keys, and cross-checked outcome metrics agree with embedded summaries. Explicit-user-input flags distinguish user submissions from system-mediated context: Liqo contains 161 typed and 136 speech submissions, and VR contains 152 typed and 164 speech submissions.

The 78 spatial files have increasing timestamps and compatible trajectory/validity dimensions. Median sampling interval is approximately 1/30 s; invalid-head flags cover 0.21–0.89% of recorded frames, in addition to omitted samples. Coordinates are Y-up with xyzw quaternions and origin/recentering fields. Head yaw is a body-forward proxy, not measured torso orientation. Revisit outcomes use target-linked readable-access events, not quadrant occupancy.

D.4 Coding and Artifact Assessment

The formal codebooks were prepared and calibrated before analysis. A7 and B7 first completed nine practice and 18 calibration cases independently. Four calibration cases produced an initial disagreement: one constraint case, two retention cases, and one revision case. Discussion was limited to these training cases. The resulting clarifications separated indoor status from venue completeness, semantic correspondence from source traceability, and revision validity from nondestructiveness. Both coders then independently rechecked all 27 training cases. Version 1.0 of CP7, CON7, RET7, and SEG7 was frozen before formal coding; no formal-period amendment or case consultation occurred, and training cases were excluded from analysis.

C7 generated the formal packets before coding. Both coders received the same packet contents in different orders, under opaque case and packet identifiers. Packets omitted condition and interface names, participant and condition-bearing session identifiers, condition and task order, native operation names, interface media, prior judgments, resolved outcomes, and statistical results. Place names were standardized. CON7 packets showed task requirements and complete structured fact snapshots; RET7 packets showed the source fact, selected unit, same-key final candidates, and opaque lineage tokens; SEG7 packets showed before/after fact snapshots, condition-neutral semantic actions, and relative times. Coders knew the general assessment purpose but not the directional hypotheses or condition mapping. These controls establish masking of explicit condition information, although content or behavior patterns could still permit inference.

A7 coded from August 10–31 across 32 sessions (85.3 active hours); B7 coded from August 11–September 2 across 34 sessions (91.5 active hours). Access controls restricted each coder to their own assignments, packets, and judgments. Both independently locked their files before C7 merged them. D7 had been designated before formal coding to review discrepancies against the frozen rules and identical masked evidence without overwriting either original label. There were no formal disagreements, so D7 performed agreement confirmation rather than substantive adjudication. The resolved results were locked before the condition mapping was released and before statistical comparison began. Table D.3 summarizes the formal targets.

Table D.3. Formal human-coding targets. Each target carries separate A7 and B7 labels; every paired label agreed. Whole-fact continuity supports the granularity sensitivity analysis, whereas atomic continuity is the primary outcome.

Scope	Codebook	Separate labels	Targets
Constraint state	CON7	Specified criterion passes	20,630
Whole-fact continuity	RET7	Eligible; same-key semantic match; traceable	2,935
Atomic continuity	RET7	Eligible; same-key semantic match; traceable	4,018
Revision segment	SEG7	Valid revision; nondestructive	819
Total			28,402

Each CON7 target retains the task criterion and structured evidence. Each RET7 target retains its source range, same-key before and after text, applicability reason, final-node reference, and semantic-input SHA-256 value. The primary continuity analysis uses 4,018 atomic clause or value spans; 2,935 whole structured facts support the granularity

sensitivity analysis. RET7 requires separate eligibility, semantic-match, and traceability labels, but its candidate-generation rule does not expose content under a different canonical key. SEG7 groups edits with the same intent separated by at most 30 s; a valid revision improves a visible constraint or adds an explicit rationale, while a nondestructive revision also preserves eligible atomic content and lineage without delete/recreate actions. Its 819 targets retain event links, before/after state identifiers, constraint gains, retained-unit counts, and destructive-action flags. Complete agreement is descriptive: repeated state judgments within sessions remain dependent, and agreement does not externally verify the truth of the semantic judgments.

Artifact assessment used a separate process. Both artifact raters used shared 1–7 anchors across the four dimensions. A score of 1 denotes major omissions or contradictions; 4 denotes a partly usable plan with multiple unresolved requirements; and 7 denotes a complete, coherent, executable plan with explicit tradeoffs and satisfied constraints. These are shared anchors, not separately validated dimension-specific rubrics. The analyzed dimension score is the average of the two independent ratings. ICC(A,2) estimates and participant-cluster 95% bootstrap intervals are .889 [.837, .921] for coherence, .883 [.775, .936] for coverage, .921 [.795, .961] for constraint satisfaction, and .937 [.919, .951] for actionability.

D.5 Same-Key Continuity Denominators and Content Examples

The primary unit is one clause or value span within a structured fact. Delimiters and recorded character ranges split compound schedules into separately assessable activities while leaving indivisible values, such as a walking limit, as one unit. Eligibility excludes changed numerical limits, old budget allocations, and superseded activities; it retains unaffected execution information and alternatives still useful for comparison. We also recompute continuity with one whole structured fact per unit. Table D.4 reports both definitions alongside execution coverage at the change point. Starting execution coverage differs between conditions and is not required for inclusion, so the revision comparison is not restricted to initially complete plans.

RET7 supplies semantic coders with final candidates sharing the source unit’s canonical key. The reported measure therefore asks whether applicable content and its lineage continue under that key. It cannot identify a paraphrase, merge, or migration represented only under a different key. Within the issued candidates, semantic-match and traceability labels coincide for every eligible unit, making the semantic, traceability, and joint percentages numerically identical. We retain the separate definitions because they remain conceptually distinct, but do not infer whole-artifact semantic retention from these data.

Table D.4. Recorded starting execution coverage and same-key continuity under atomic and whole-fact units, across 39 trials per condition. Rates average task ratios within participants; pooled continued/eligible counts therefore need not yield the reported percentages.

Condition	Initial execution	Atomic eligible /trial [range]	Atomic continued/eligible	Atomic mean (%)	Whole-fact continued/eligible	Whole-fact mean (%)
Chat	18/39	16.85 [7–25]	604/657	91.84	358/387	91.83
Canvas	25/39	16.44 [10–24]	604/641	94.32	347/371	93.27
VR baseline	27/39	16.85 [8–27]	602/657	91.18	347/386	89.63
Liqo	20/39	16.92 [10–26]	658/660	99.74	386/388	99.60

P01’s Liqo travel trial contains 34 initial atomic units, of which 26 are eligible and all 26 continue under their keys. The walking fact changes from 6 to 3 km and remains linked to node N008, but the old value is ineligible because it

is superseded. The second-day itinerary is split into four activity spans: the morning workshop, lunch, and evening teahouse remain eligible and continue, while the outdoor afternoon market is ineligible after the task requires an indoor activity. Under the whole-fact rule, the compound itinerary is excluded because not all of its clauses remain applicable. The atomic analysis therefore preserves evidence about unchanged subparts without treating the obsolete activity as a loss. The accommodation alternative N011 also retains the translated content “Riverside-view lodging, more expensive but convenient for night views” and remains linked to N011 in the final artifact.

P08’s Liqo travel trial supplies a same-key loss case: its eligible accommodation alternative, “Riverside-view lodging, more expensive but convenient for night views,” has neither a semantic match among the same-key candidates nor a traceable final node. P13’s Liqo market trial similarly loses the eligible contingency “Use an indoor waiting area during rain or congestion” under its key. These are the two discontinuities among Liqo’s 660 eligible atomic units. The final task criteria can pass despite such a loss because critical success and structured continuity answer different questions. The candidate scope does not establish absence from every differently keyed final node, and the records do not identify which transformation caused either discontinuity.

D.6 Failure Categories and Recovery

In VR, final travel failures involve execution coverage (2 trials) and walking (1); event failures involve execution (7), meal quantity (2), indoor venue (2), capacity (1), vegetarian meals (1), and accessible entry (1); market failures involve execution (4) and the limited-trial decision (1). Categories overlap within trials. Liqo’s sole final failure is a travel budget overrun. This contrast locates the observed task heterogeneity without attributing it to a single interface feature.

For Liqo’s 71 native-operation target failures, 31 have successful linked recovery, 39 have no linked recovery attempt, and one has an attempt without a successful endpoint. All 40 without success are marked right-censored through 1200 s. P12’s event-task operation FORMAL_OP015, at 935.5–938.7 s, targets two children containing team-activity and catering-cost facts. Its observed state retains their parent and both fact keys instead. A later linked attempt, FORMAL_OP025, has no successful recovery recorded in the remaining 261.3 s. Thus the recorded failure concerns the intended structure, not evidence of semantic deletion. No linked attempt cannot distinguish abandonment, continued work elsewhere, or incomplete recovery linkage.

D.7 Operation-Expectation Probe Composition

Table D.5 gives the actual VR probe controls and counts. Each trial contains a subset and frequency of probes; the main analysis first averages within trial, then across three tasks per participant. Native formal operations and these standardized probes have separate scopes. In Chat, the same five semantic categories map to combine sections, split outline, extract subsection, save version, and reopen version; in Canvas they map to merge, split, extract, pin, and unpin cards. The records confirm a tutorial and practice task for each interface but do not retain per-control exposure or practice counts. These mappings therefore describe prediction for the tested controls, not functional equivalence during planning or intrinsic metaphor intuitiveness.

D.8 Reading-Panel Allocation and Presentation

The reading probe uses 12 materials: four surface variants associated with each of the three task identifiers. Every passage has 143 Chinese characters and instantiates the same ten fact types, including two budgets, two staff/time associations, three schedule times, an indoor location, an accessible route, an emergency location, and a reserve amount. Within each immersive condition, every material occurs in three or four sessions. All 780 immersive question records

Table D.5. VR post-task prediction matches/counts by semantic category. Each response predicts object count and editability. Counts are descriptive and are not independent participant observations.

Category	Liqo control; matches/count	VR control	VR matches/count
Fusion	Fuse; 63/66	Group objects	46/60
Fission	Split; 63/68	Ungroup objects	44/58
Exudation	Exude; 67/69	Duplicate/extract	55/68
Freeze	Freeze; 64/69	Lock object	56/66
Melt	Melt; 73/77	Unlock object	50/63

use 18-point text and 0.6° angular character height at 1.1 m; the ten questions appear in varying orders. The variants were not independently normed for equal difficulty.

Reading accuracy averaged 93.8% in Liqo and 80.5% in VR (difference 13.3 points, 95% CI [10.8, 15.9]); mean question-response time was 8.16 s versus 11.24 s (difference -3.08 s, CI $[-3.50, -2.66]$; both $p_A = .0056$). These secondary outcomes remain in the joint 23-contrast adjustment. They concern the recorded panels, not general reading ability or sustained comprehension of a participant's plan.

D.9 Questionnaire Records

Each of the 156 sessions contains one questionnaire submission with a unique submission identifier, form identifier and version (Q7, version 1.0), opening and submission timestamps, and nine item-response records. All 1,404 response identifiers are unique. Every item retains its Chinese prompt, numeric range, endpoint anchors, selected value, presentation and answer timestamps, and input mode. The corresponding session summary points back to the same submission and response identifiers. All sessions pass cross-checks between these item records and the analyzed values.

The six Raw NASA-TLX items are mental demand, physical demand, temporal demand, performance, effort, and frustration on 0–100 response ranges. The selected performance value is transformed to performance failure before the unweighted mean, giving every component the same higher-is-worse direction. The remaining items measure interface comfort, readability, and physical discomfort on 1–7 scales. Section 5 reports their wording and anchors in translation. There are no missing questionnaire items; the analysis supplement retains all prompts, anchors, identifiers, values, and timestamps.

D.10 Recorded Travel Revision

Table D.6 details P01’s travel revision. The prior request R003 completed at 508.8 s, before freezing began. R004 begins when melting completes, but its recorded trigger is the explicit spoken submission. The final plan lists CNY 4,019 against a revised budget of CNY 4,160. Because comparative-study video and screenshots were not recorded, the account establishes these state relations without reconstructing the participant’s rendered view or reasoning.

Table D.6. P01’s recorded travel revision. Content entries translate structured facts; times are seconds from task start. This reconstruction links logged events and states, not screenshots or verbatim speech.

Time	Event / state	Content evidence
511.1	S004, before freeze	N008: daily walking limit 6 km.
517.7–519.7	Freeze	N008 becomes solid with the same walking-fact key.
600.0; 623.8	Constraint change; spoken input I004	Revised limit 3 km; request to reorganize the route and compare with the earlier plan.
633.9–636.3	Melt	N008 reopens with the same identifier and fact key.
636.3–658.7	Request R004	Model exchange triggered by I004.
661.9; 687.6	S005; S009	Walking limit remains 6 km, then changes to 3 km.
697.5	S012	First recorded state passing all critical criteria.

D.11 Complete Paired Results

All contrasts below are Ligo minus VR baseline using 13 participant means. The final column adjusts all 23 tests jointly and supplies every main-text p_A value. The p_F column gives the alternative within-family adjustment for the five exploratory families in Section 5. Raw values, task/order allocation, bootstrap settings, source manifests, request prompts, and audit outputs are retained with the analysis supplement. Paired-difference intervals use 20,000 bootstrap samples; ICC intervals use 4000 participant-cluster samples over the 156 paired artifact assessments. The analysis seed is 20260909, distinct from the recorded model-request seed.

Table D.7. Exploratory Ligo minus VR contrasts, $n = 13$ paired participant means. VR and Ligo columns are means; intervals are unadjusted participant-bootstrap 95% CIs. Percentage-valued outcomes have percentage-point differences. p_F : Holm adjustment within the named family; p_A : Holm adjustment across all 23 tests. CIs are not simultaneous and need not agree with adjusted tests.

Outcome	VR	Ligo	Difference [95% CI]	p_F	p_A
<i>Revision Core</i>					
Final success (%)	61.54	97.44	+35.90 [25.64, 46.15]	0.0010	0.0107
Restricted revision time (s)	434.86	186.57	-248.30 [-281.82, -211.60]	0.0007	0.0056
Same-key continuity (%)	91.18	99.74	+8.56 [5.42, 12.13]	0.0007	0.0056
<i>Artifact Quality</i>					
Coherence	5.87	6.46	+0.59 [0.31, 0.87]	0.0039	0.0205
Coverage	6.17	6.79	+0.63 [0.40, 0.87]	0.0029	0.0107
Constraint rating	6.44	6.88	+0.45 [0.26, 0.67]	0.0039	0.0137
Actionability	5.49	6.46	+0.97 [0.68, 1.29]	0.0010	0.0056
<i>Process</i>					
Hierarchy depth	3.97	4.15	+0.18 [0.03, 0.33]	0.1875	0.2813
Active branches	1.59	1.34	-0.25 [-0.46, -0.02]	0.1670	0.2227
Deleted units	1.85	0.21	-1.64 [-2.00, -1.28]	0.0017	0.0056
Recreated units	0.79	0.15	-0.64 [-1.00, -0.31]	0.0234	0.0293
Repeated checks	8.28	6.95	-1.33 [-2.64, 0.13]	0.1875	0.2813
Nondestructive revision (%)	63.97	94.66	+30.68 [18.38, 42.44]	0.0073	0.0117
Expectation match (%)	79.35	94.77	+15.42 [8.91, 22.64]	0.0029	0.0063
<i>Spatial And Reading</i>					
Revisit success (%)	58.97	100.00	+41.03 [28.21, 56.41]	0.0010	0.0107
Restricted revisit time (s)	24.68	6.67	-18.01 [-21.95, -14.31]	0.0010	0.0056
Reading accuracy (%)	80.51	93.85	+13.33 [10.77, 15.90]	0.0010	0.0056
Reading response time (s)	11.24	8.16	-3.08 [-3.50, -2.66]	0.0010	0.0056
<i>Experience</i>					
Interface comfort	4.28	5.77	+1.49 [1.15, 1.82]	0.0012	0.0056
Readability	4.92	6.05	+1.13 [0.90, 1.36]	0.0012	0.0056
Discomfort	1.92	1.79	-0.13 [-0.38, 0.18]	0.5176	0.5176
Raw NASA-TLX	57.07	38.33	-18.74 [-24.10, -13.25]	0.0012	0.0063
Physical demand	50.38	31.54	-18.85 [-24.36, -13.08]	0.0012	0.0056